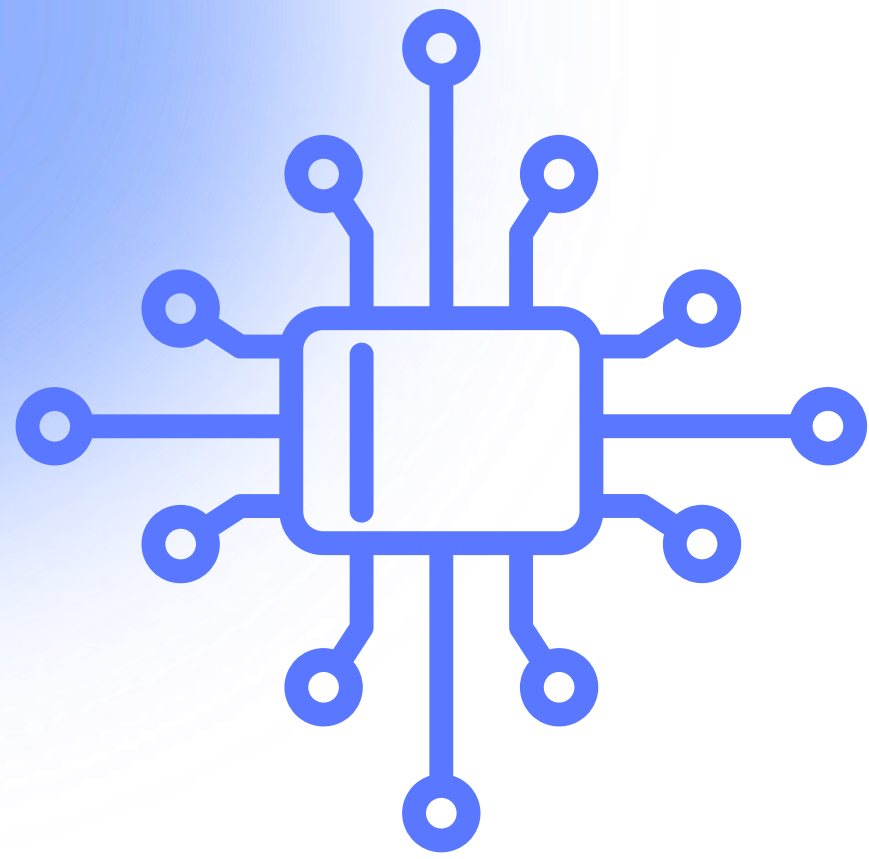
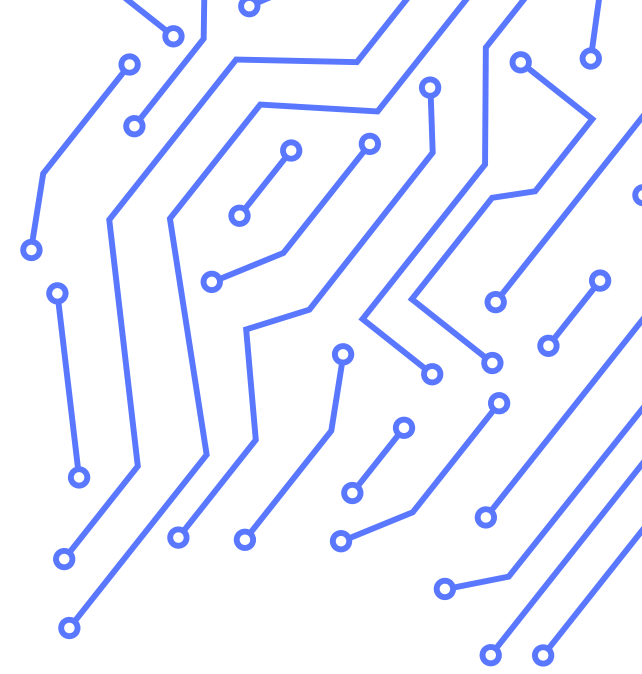
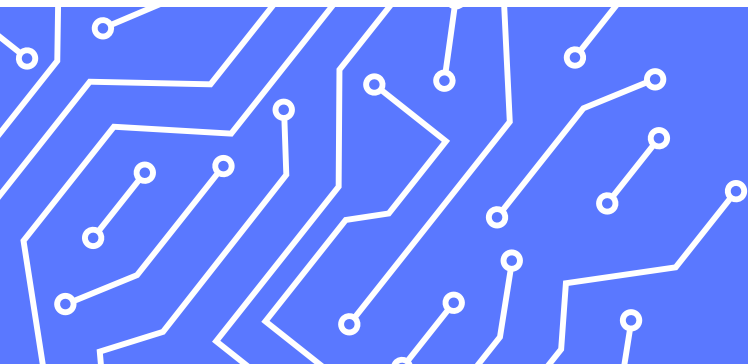




NEURONS AND LAYERS



**MADE BY:
SANDY ALAA**

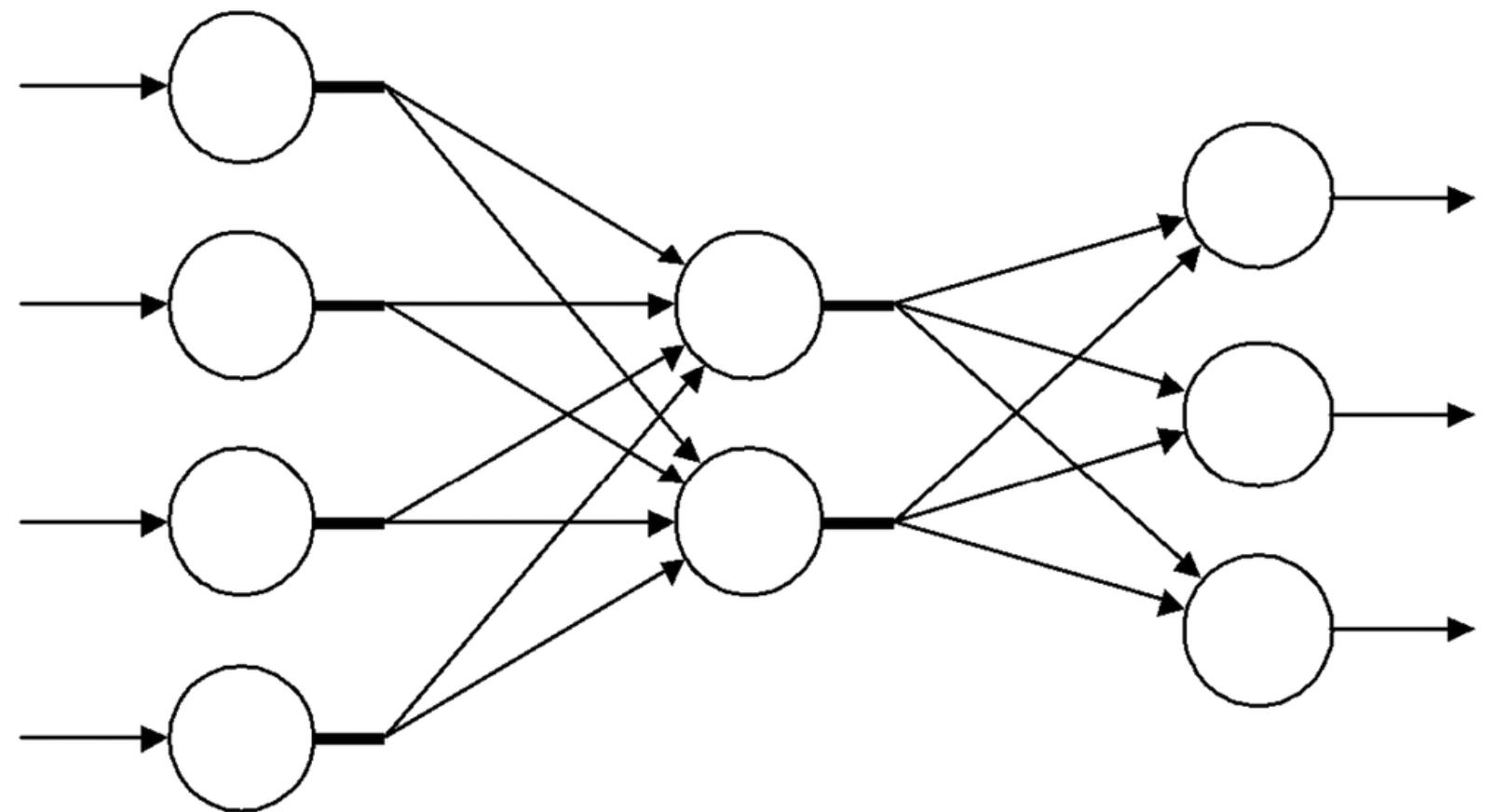


INTRODUCTION

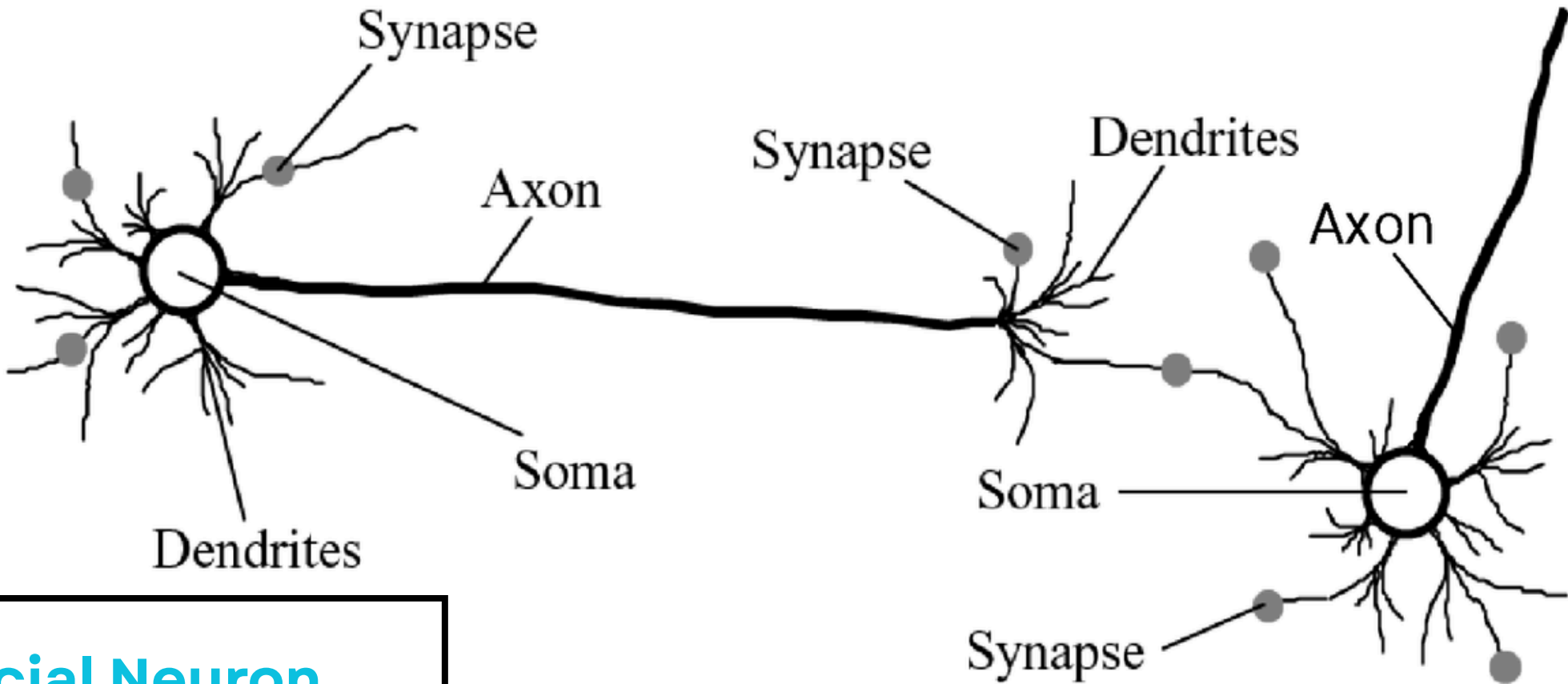
Before anything, we need to understand its most basic **building blocks**:

Neuron → Layer → Neural Network

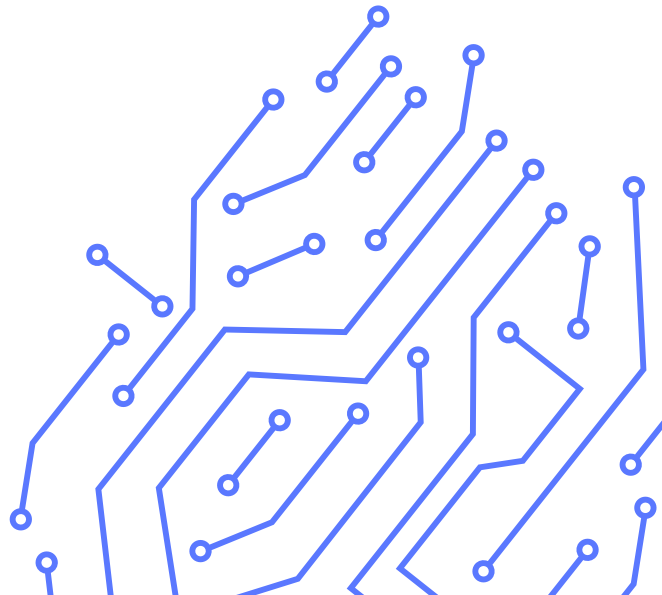
Neuron: is a mathematical computational unit that receives one or more inputs and combines them to produce an output. It is also commonly called: **Node, Unit and Neuron**



BIOLOGICAL VS ARTIFICIAL NEURON



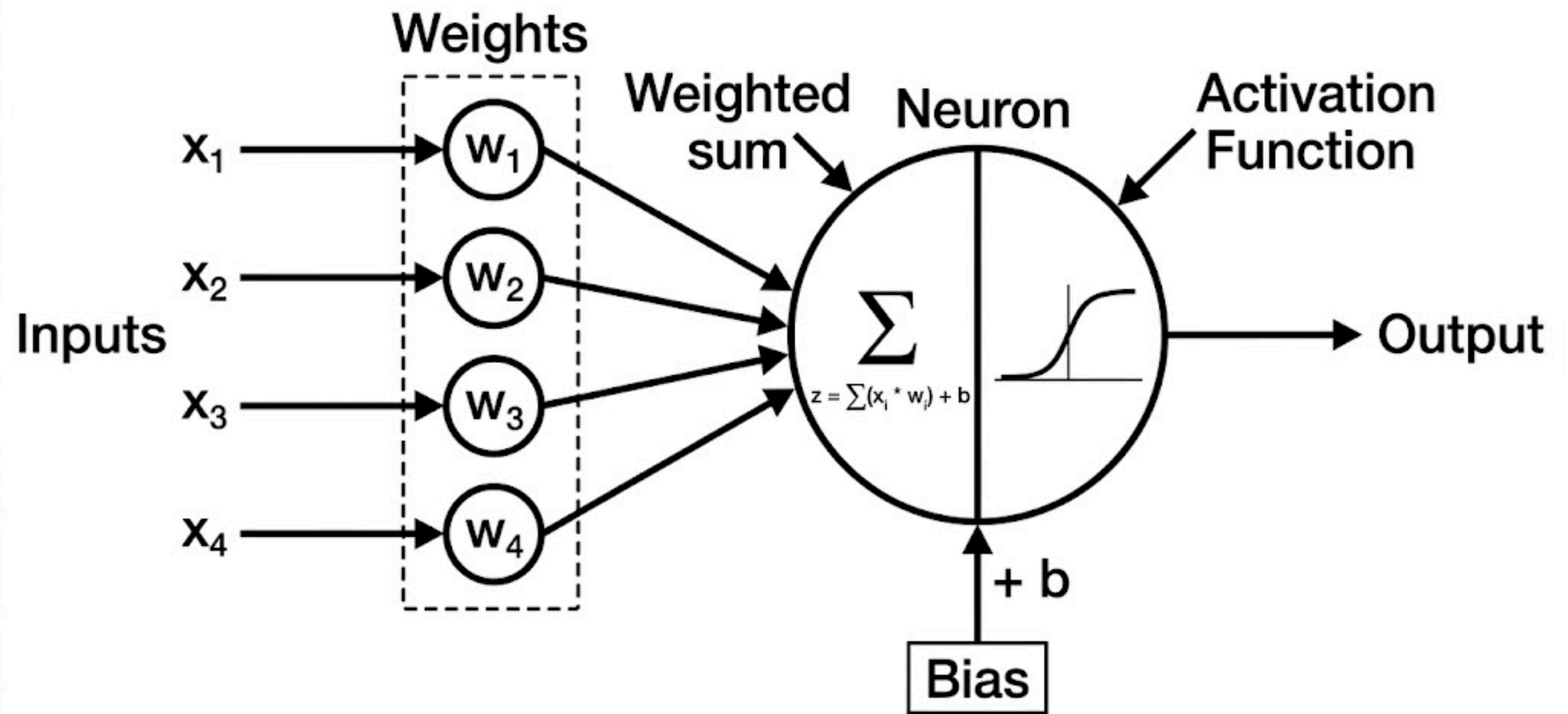
Biological Neuron	Artificial Neuron
Dendrites receive signals	Inputs
Synapses control signal strength	Weights
Cell body processes signals	Mathematical computation
Axon carries the signal	Output



ANATOMY OF NEURON

THE MAIN COMPONENTS:

- Inputs
- Weights
- Bias
- Weighted sum
- Output



ANATOMY OF NEURON

Inputs: are the numerical values provided to a neuron. We can represent them as:

- A feature from a dataset
- A numerical measurement
- A pixel value
- Or the output of a neuron from a previous layer

Weights: determines how strongly that input contributes to the neuron's weighted sum.

Bias: is an additional parameter associated with a neuron. It provides an adjustable offset

It is added to the weighted sum z :

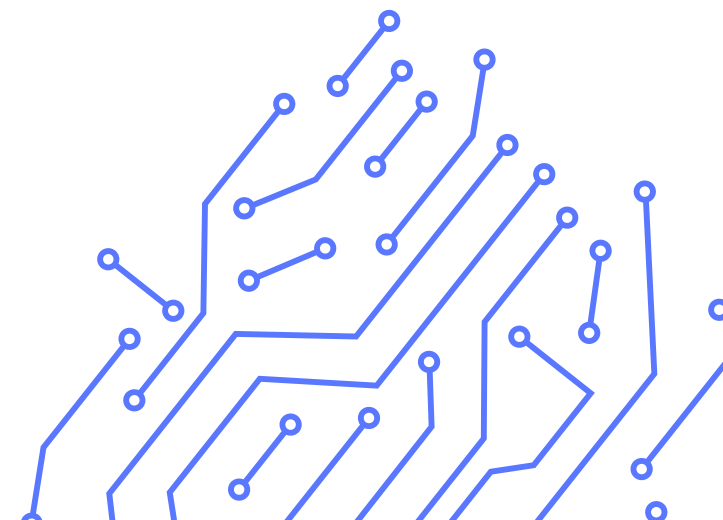
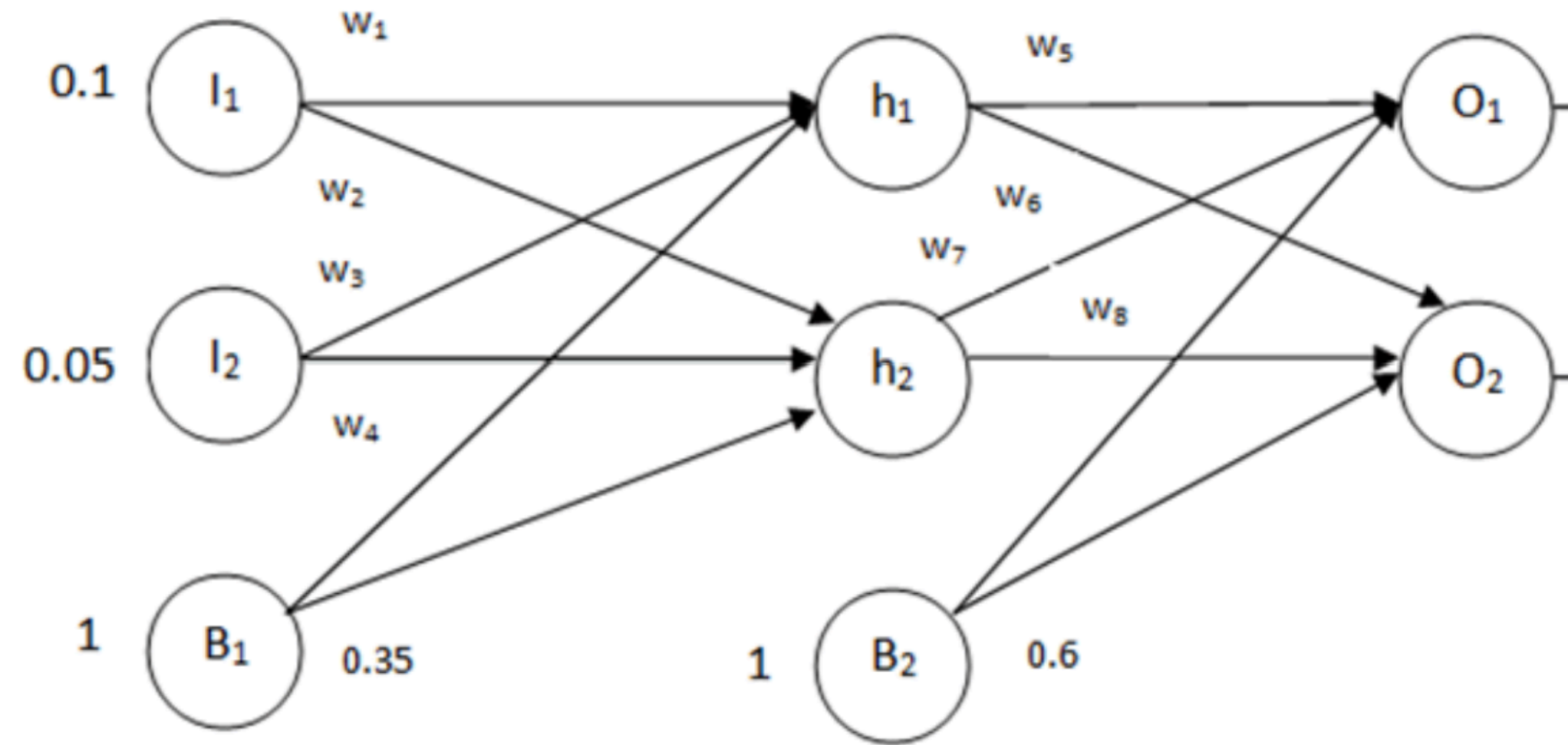
$$z = w_1x_1 + w_2x_2 + \dots + w_nx_n + b$$

$$z = \sum_{i=1}^n w_i x_i + b$$



EXAMPLE

- What will be the values of h_1 and h_2 , given that $w_1 = 0.2$, $w_2 = 0.15$, $w_3 = 0.3$, $w_4 = 0.25$, $w_5 = 0.45$, $w_6 = 0.4$, $w_7 = 0.55$, $w_8 = 0.5$? (assume bias weights equal 1)
- Using the results in (1), what will be the values of o_1 and o_2 given that the $w_5 = 0.45$, $w_6 = 0.4$, $w_7 = 0.55$, $w_8 = 0.5$?



VECTOR REPRESENTATION

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad \mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}$$

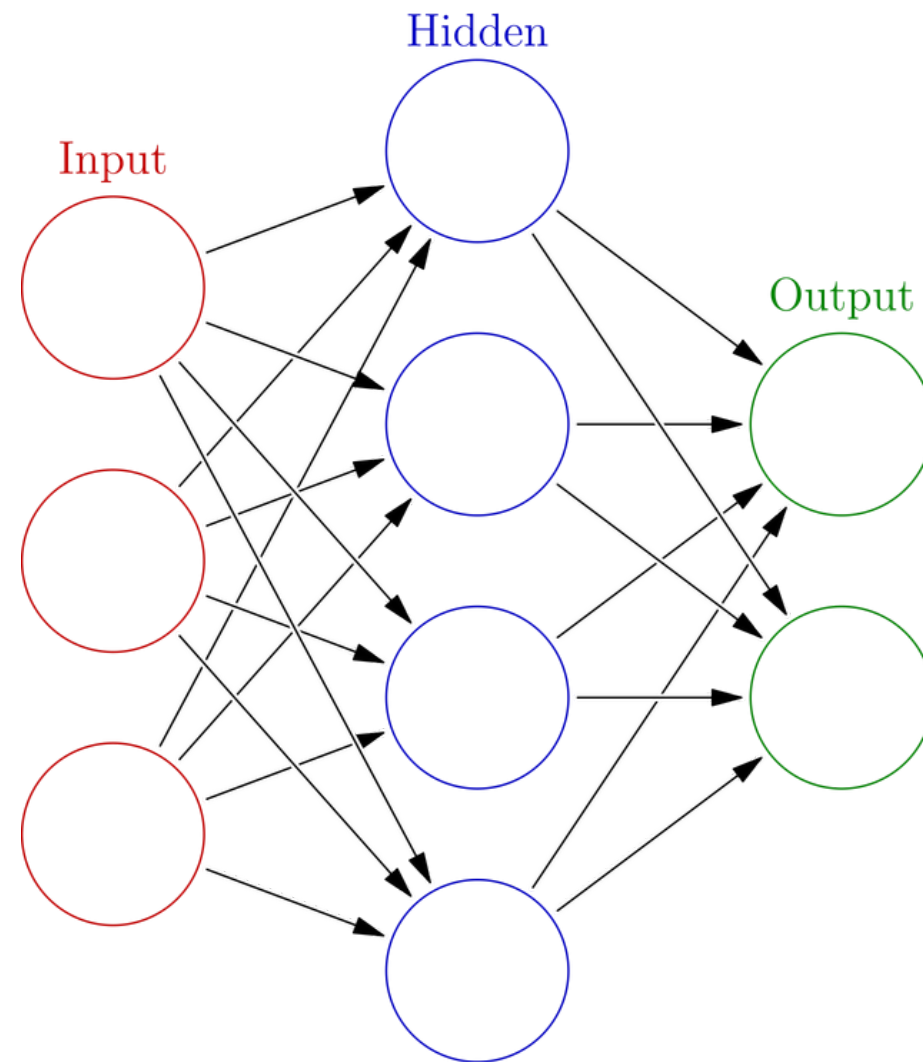
$$z = \mathbf{w}^T \mathbf{x} + b$$



LAYERS

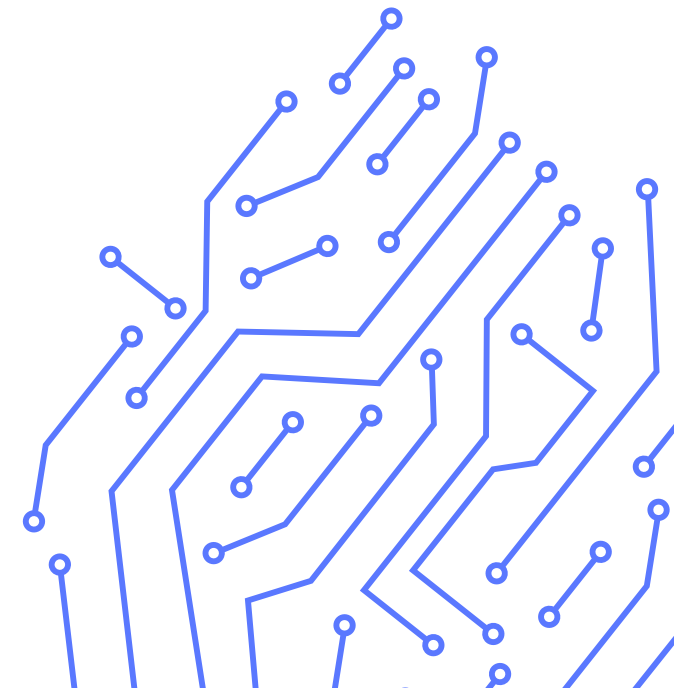
Layer: is a group of neurons/units operating at the same stage of a neural network.

Types of layers:



Hidden Layers:

Each layer receives information from the previous stage and produces a new representation for the next stage.



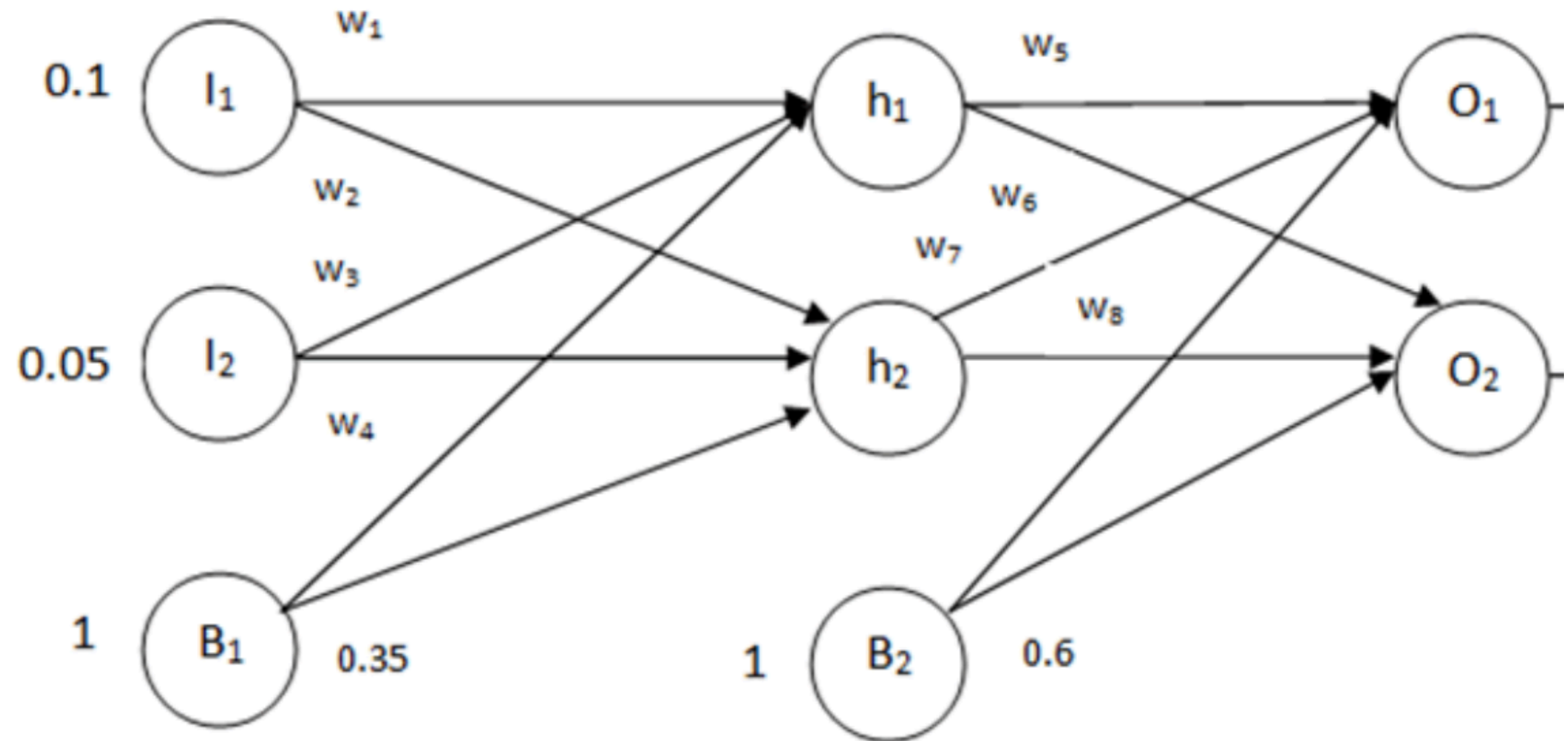
LAYERS

Dense / Fully Connected Layers: connects every unit from the previous layer to every neuron in the current layer.

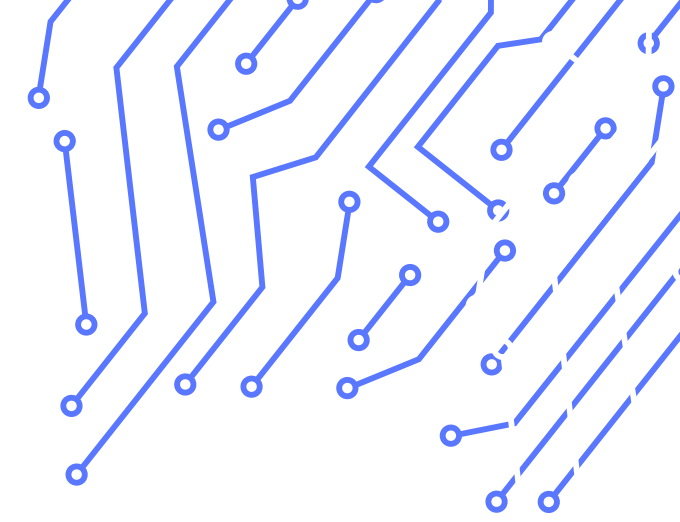
Counting Connections:

- **Connections** = Weights
- **Parameters** = Weights + Biases

Calculate the connections and parameters



QUESTIONS



Question 1

A neuron receives 5 inputs.
How many weights does it have?

Answer:

Question 2

A layer contains 20 neurons, each receiving 10 inputs.
How many parameters?

Answer:

Question 3

What is the difference between width and depth?

Answer:

Question 4

If Layer A contains 5 neurons and Layer B contains 8 neurons, how many connections exist in a fully connected architecture?

Answer:

THANK YOU

I hope you understand the
session well

Thanks for your time

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